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Response to the Reviewer

We are grateful to the reviewer for the detailed comments, which are very valuable. We have revised the manuscript based on the comments. In what follows, we state our corrections point by point.

1 & 2: The local linking argument does not cover $p = 2$ (Lemma 2.1, p. 5); Make the admissible range of p explicit.

For the validity of Lemma 2.1 we have to require $p > 2$. Also, for the term $\int |\nabla u|^p < +\infty$, we need $p \in [2, 2^*]$. Hence the range of p should be $p \in (2, 2^*]$. We have added the explicit range of p in the theorems and in Lemma 2.1.

A paragraph concerning the case $p = 2$ is also added right after Theorem 1.2.

3: Clarify the spectral meaning of (V_0).

Yes, the nondegeneracy in (V_0) means that this representing operator is an isomorphism. We have specified this in the statement of (V_0).

4: Complete several proof details.

We have revised the proof of Lemma 2.3. Using argument based on subsequences. In particular, the well known fact that "if each subsequence of $\{a_n\} \subset \mathbb{R}$ has a subsequence converging to a, then $a_n \to a$" is used to prove $\langle \mathcal{N}(u_n), \phi \rangle \to \langle \mathcal{N}(u), \phi \rangle$.

In the original proof of Proposition 4.1, the problem is that $2_* = \infty$ when $N \leq 4$. Then the Hölder inequality would not be true. To fix this problem, as suggested by the reviewer, we introduce a finite exponent r_i, which equals 2_* if $N \geq 5$. Then replacing the problematic 2_* with r_i. In this way the argument works for both $N \geq 5$ and $N \leq 4$.

We start the proof of Theorem 1.2 with a short paragraph justifying to assume that $u = 0$ is the only critical point of Φ. This will also imply that critical values of Φ is bounded below by $-A$, so that

$$C_k(\Phi, \infty) = H_k(X, \Phi_{-A});$$

we don't need to take a bigger A to ensure $\Phi(\mathcal{K}) \subset (-A, \infty)$, where $\mathcal{K}$ is the set of all critical points.

A homotopy H yielding $\Phi_{-A} \simeq X \setminus B_1$ is also presented, justifying the isomorphism in (3.2).

5: Corrections and misprints.

We have corrected all the misprints as listed. In particular, condition (f_0) in Lemma 2.5 and (1.7) in Lemma 2.6 have been corrected into (f_1). The tail estimate in Remark 4.2 is also corrected using the exponent r_i.